MILITARY SPECIFICATION

SENSOR GROUP, ULTRASONIC MOTION: TRANSCEIVER

RT-1161/FSS-9(V) AND PROCESSOR MX-9444/FSS-9(V)

This interim amendment is approved for use by U.S. Army Belvoir R&D Center with Military Specification MIL-S-52872B, dated 24 March 1980.

PAGE 1

2.1, lines 1-3, delete and substitute:

"2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein."

PAGE 2

2.1, under SPECIFICATIONS, MILITARY, delete:

"MIL-C-45662 - Calibration System Requirements."

2.1, under STANDARDS, MILITARY, delete:

"MIL-STD-1188 - Commercial Packaging of Supplies and Equipment."

2.1, under STANDARDS, MILITARY, add:

"MIL-STD-45662 - Calibration System Requirements."

After MIL-STD-810, add "1/"

2.1, add the following note at bottom of page:

"1/ The 'C' revision of MIL-STD-810 shall be used."

2.1, immediately above DRAWINGS, insert:

"2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein."

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2.1, immediately following parenthetical statement ending with "contracting officer.)" add:

"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Standard Practices for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)"

3.1.1, line 3, delete item (a) and substitute:

"(a) Transceiver, Ultrasonic Motion Signal TA 13220E3226".

Line 5, delete item (c) and substitute:

"(c) Processor, Ultrasonic Motion Signal TA 13220E3201."

PAGE 10

3.6.2.4, line 2, delete "detector" and substitute "transceiver".

PAGE 11

3.6.2.6, line 3, change to read: "the following characteristics when tested as specified in test Method 7 of the Appendix."

3.6.2.6 (a), delete and substitute:

"(a) Operating voltage and current criteria. With the signal generator turned off and +20 ± 0.1 Vdc applied to TB1-7, referenced to TB1-8, the following conditions shall exist:

(1) An input current to TB1-7 of 35 ±5 milliamperes DC.
(2) A voltage of 12 ±0.6 Vdc at the plus side of capacitor C35 referenced to TB1-8.
(3) A voltage of 6.8 ±0.6 Vdc between the reference test point and TB1-8.
(4) A voltage of ±0.2 Vdc between the balance and reference test points.
(5) For a voltage reversal input, TB1-7 negative and TB1-8 positive, the reverse current shall not exceed 1 milliampere."

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3.6.2.6 (b), delete and substitute:

"(b) Alarm criteria for Doppler signals.

(1) An AC signal of 26.3 KHz ±130 Hz at 5.7 ±0.2 Vrms shall exist between TB2-1 referenced to signal ground TB2-4.
(2) A doppler frequency of 400 ± 20 Hz shall exist at the NOISE test point when the signal generator is turned on and adjusted to a frequency of 400 ± 20 Hz greater than that obtained in 3.6.2.6 (b) (1) above at a voltage, as measured at E2 referenced to TB2-4, equal to 33 percent ±2 percent of that obtained in 3.6.2.6 (b) (1) above.
(3) The no-alarm state shall exist, resistance of 100 ±10 ohms between TB-1 and TB1-2, when the attenuators are set to 42db.
(4) The alarm state shall exist, resistance of not less than 100K ohms between TB-1 and TB1-2, when attenuators are set in the range of 37 ±3db."

3.6.2.6 (c), delete and substitute:

"(c) Alarm criteria for noise.

(1) When the doppler channel is deactivated (clip lead from the positive side of C43 to the junction of R52 and R53) a resistance of 100 ±10 ohms shall exist between TB-1 and TB1-2 when attenuators are set to 32 ±1 db. A resistance of not less than 100K ohms shall exist between TB1-1 and TB1-2 when attenuators are set in the range of 25 ±3db."

3.6.2.6 (d), delete and substitute:

"(d) Log amplifier gain criteria.

(1) With doppler channel deactivation clip lead removed and attenuators set to obtain 1.0 ±0.1 Vrms at the SIG test point, a reduction in attenuation of 20db shall result in 1.8 ±0.3 Vrms at the SIG test point."

3.6.2.6 (e), delete and substitute:

"(e) Linear amplifier gain and doppler filter criteria.

(1) When attenuators are set to 20db, a voltage of 2.8 ±0.3 Vrms shall exist between the NOISE test point and TB2-4. For an attenuation of 40db the voltage at the NOISE test point shall be reduced by a factor of 10 ±10 percent. A decrease in the NOISE test point

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voltage of at least 3db shall be obtained when the signal generator is set to obtain a frequency at the NOISE test point of +800 ±20Hz, -800 ±20Hz, ±12Hz, +2Hz and -12 ±2Hz."

3.6.2.6 (f), delete and substitute:

"(f) Gating waveform criteria. A pulse of +12 ±1 volts amplitude and 100 ±35 microseconds in duration shall exist at the junction of resistors R51 and R57. When the signal generator frequency is set in accordance with 3.6.2.6(b)(2) the pulse shall occur at the negative peak of the SIG test point signal. Readjustment of the signal generator to 400 ±20 Hz less than that obtained in 3.6.2.6(b)(1) shall result in the pulse occuring at the positive peak of the SIG test point signal."

3.6.2.6 (g), delete and substitute:

"(g) Transmitter failure alarm test. When attenuators are set to 70db a resistance of 100 ±10 ohms shall exist between TB1-1 and TB1-2. A resistance of at least 100K between TB1-1 and TB1-2 shall exist when the frequency control transducer red lead is disconnected for any duration above 250 milliseconds. The restoration of the transducer red lead connection shall result in a resistance between TB1-1 and TB1-2 of 100 ±10 ohm within 1.5 seconds."

3.6.2.6 (h), delete and substitute:

"(h) Synchronization output test. A symmetrical triangular waveform of 3.0 ±0.3 volts PK to PK shall exist between TB2-7 and TB2-4."

3.6.2.6 (h), add:

"(i) Tamper switch test. A resistance of 1 ohm or less shall exist between TB1-5 and TB1-6 when tamper switch plunger is fully depressed or fully extended. A resistance between TB1-5 and TB1-6 of not less than 100K ohms shall exist after switch plunger is momentarily depressed and then released."

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after 3.6.2.9, add:

"3.6.2.10 Transceivers, ultrasonic motion signal. The transceiver, ultrasonic motion signal, shall be in accordance with top assembly TA 13220E3226, and its component parts shall conform to 3.6.2.4, 3.6.2.5, 3.6.2.7 (b) and (c), 3.6.2.8 and 3.6.2.9."

"3.6.2.11 Processor, Ultrasonic Motion Signal. The processor, ultrasonic motion signal shall be in accordance with top assembly TA13220E3201, and as specified in 3.6.2.4 and 3.6.2.6."

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PAGE 16

4.1.3, delete in its entirety.

PAGE 17

4.3.2.2, line 4, delete "VII" and substitute "IX".

PAGE 18

4.5.3.1.2, line 3, delete "VII" and substitute "IX".

4.5.3.2.2, line 3, delete "VII" and substitute "IX".

4.6.1, defect 101, delete in its entirety.

PAGE 19

4.6.2.1, line 3 after "conditions.", delete and substitute:

"Prior to testing, each complete UMS, and major components 3.1.1 (a), (b), (c) and (d) when furnished separately, shall be subjected to a 72-hour burn-in. During burn-in, the test items shall be continuously energized, and shall be exercised nine times during each 24 hours at approximately equal intervals as follows:

a. Complete UMS's: Apply signal specified in 2.1 (g) of Appendix Test Method 2.
b. Transceiver and transceiver circuit card assembly: Perform steps 2.1(c) and 2.1(d) of Appendix Test Method 6.
c. Processor and processor circuit card assembly: Perform step 2.c(3) of Appendix Test Method 7.

Circuit card assemblies need not be placed in enclosures to simulate their normal environment for this test. All failures shall be recorded and analyzed for cause of failure. Corrective action shall be taken when appropriate. Copies of these failure reports shall be forwarded to the contracting officer. Items for testing shall be selected from those units having successfully completed the 72-hour burn-in."

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After Table VII, add Tables VIII and IX:

TABLE VIII. Test schedule for transceiver.

Production	Quality conformance: Individual	Quality conformance: Sample	Tests	Requirement Paragraph	Test Paragraph
1	2	3	4	5	6
X	X	-	Functional	3.6.2.10	4.6.2.4.10
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

TABLE IX. Test schedule for processor.

Production	Quality conformance: Individual	Quality conformance: Sample	Tests	Requirement Paragraph	Test Paragraph
1	2	3	4	5	6
X	X	-	Functional	3.6.2.11	4.6.2.4.11
X	-	X	Low temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.1
X	-	X	High temperature	3.5.1(a), 3.5.2(a)	4.6.2.5.2
X	-	X	Humidity	3.5.1(b), 3.5.2(b)	4.6.2.5.3
X	-	-	Shock	3.5.2(c)	4.6.2.5.4
X	-	-	Vibration	3.5.2(d)	4.6.2.5.5

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4.6.2.3, line 3, delete "MIL-C-45662" and substitute "MIL-STD-45662".

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After 4.6.2.4.9, add:

"4.6.2.4.10 Transceiver, Ultrasonic Motion Signal. The transceiver shall be tested in accordance with Appendix Test Method 9. Nonconformance to 3.6.2.10 shall constitute failure of this test."

"4.6.2.4.11 Processor, Ultrasonic Motion Signal. The processor shall be tested in accordance with Appendix Test Method 10. Nonconformance to 3.6.2.11 shall constitute failure of this test."

4.6.2.5, delete in its entirety and substitute:

"4.6.2.5 Environmental. The UMS and major components or transducer kits furnished separately shall be subjected to the following series of tests in accordance with MIL-STD-810C. Prior to and following each test, conduct tests as follows on complete UMS's and major components furnished separately.

a. Complete UMS's: perform operational tests in accordance with Appendix Test Method 5.
b. Transceiver circuit card assembly: Perform tests of paragraph 2 of Appendix Test Method 6.
c. Processor circuit card assembly: Perform tests of paragraph 2 of Appendix Test Method 7.
d. Transceiver: Perform tests of paragraph 2 of Appendix Test Method 9.
e. Processor: Perform tests of paragraph 2 of Appendix Test Method 10.
f. Transducers: Perform tests of applicable portion of paragraph 2 of Appendix Test Method 8."

4.6.2.5.1, delete all after the first sentence and substitute: "For complete UMS's, at the operating temperature, perform the operating test in accordance with Appendix Test Method 5. The frequency measurement shall be within ±131 Hz of the recorded reading of Step 2(e), Appendix Test Method 5, measured at room temperature (+75° F ±5° F) prior to this environmental test. For major components furnished separately, perform tests specified in 4.6.2.5 at operating temperature."

PAGE 27

4.6.2.5.2, delete all after the first sentence and substitute: "For complete UMS's, at the operating temperature, perform the operating test in accordance with Appendix Test Method 5. The frequency measurement shall be within ±131 Hz of the recorded reading of Step 2(e), Appendix Test Method 5, measurement at room temperature (+75° F ±5° F) prior to this environmental test. For major components furnished separately, perform tests specified in 4.6.2.5 at the operating temperature."

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4.6.2.5.3 (a), after the first sentence, delete and substitute: "at the completion of each cycle, test complete UMS's in accordance with Appendix Test Method 5, and test major components and transducers furnished separately as specified in 4.6.2.5."

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4.6.2.5.5, delete "curve AX" and substitute "curve AW".

PAGE 30

4.8.1.3, defect 108, delete and substitute:

"108. Preservation methods for complete UMS's and for separate major components not as specified for Levels A or B."

PAGE 32

5.1.3, line 2, delete "MIL-STD-1188" and substitute: "ASTM D3951".

PAGE 33

5.2.3, line 3, delete "MIL-STD-1188" and substitute: "ASTM D3951".

5.3.2, line 2, delete all after "in accordance with" and substitute:

"ASTM D3951 with the added requirement for weight and cube data on the shipping container. Special marking specified in 5.3.3 shall also be included."

PAGE 35

6.4.6, delete in its entirety.

6.5, delete in its entirety.

PAGES 49, 50, 51, and 52

Test Method 7: delete paragraphs 1 and 2 in their entirety and substitute:

"1. Test Apparatus:

a. Power supply, Hewlett-Packard Model 6202B or equivalent.
b. Test fixture shown in figure 2.
c. Signal generator, Hewlett-Packard Model 200CD or equivalent.
d. Oscilloscope, Tektronix Type 465 or equivalent.
e. Electronic counter, Hewlett-Packard Model 5300A/5304A or equivalent.

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f. Step attenuator, Hewlett-Packard Model 355C or equivalent.
g. Step attenuator, Hewlett-Packard Model 355D or equivalent.
h. AC Voltmeter, Hewlett-Packard Model 403B or equivalent.
i. Multimeter, Fluke Model 8020A or equivalent.

2. Test Procedure.

a. Operating voltage and current criteria.

(1) Connect the processor circuit card and test apparatus as shown in figure 2.
(2) Turn power supply on and set to +20 ±0.1 Vdc; signal generator shall be turned off. The input current to TB1-7 shall be 35 ±5 milliamperes dc. A voltage of +12 ±0.6 volts shall be measured at the plus side of capacitor C35 referenced to TB1-8. A dc voltage of 6.8 ±0.6 volts shall be measured between the REFERENCE test point and TB1-8. A dc voltage of 0 ±0.2 volts shall exist between the BALANCE and REFERENCE test points. For a voltage reversal input to TB1-7 and TB1-8 the reverse current shall not exceed 1 milliampere.

b. Alarm criteria for doppler signals.

(1) An ac signal of 26.3 Khz ±130 Hz at 5.7 ±0.2 VRMS shall exist at TB2-1 referenced to signal ground TB2-4.
(2) Connect an ac voltmeter between the signal generator output (E2) and TB2-4. Turn the signal generator on and set frequency to 400 ±20 Hz greater than that measured in 2.b.(1). Adjust the signal generator amplitude to a value equal to 33 percent ± 2 percent of that measured in 2.b.(1).
(3) Connect the counter to the NOISE test point and verify a doppler frequency of 400 ± 20 Hz.
(4) Connect the multimeter (OHMS) to terminals TB1-1 and TB1-2. With the attenuators set to 42db a resistance of 100 ±10 ohms (no alarm state) shall exist between TB1-1 and TB1-2. A resistance of at least 100K ohms (alarm state) shall occur when the attenuators are set in the range of 37 ±3 db. Allow at least 60 seconds for processor to stablize when large attenuation steps are made.

c. Alarm Criteria for Noise.

(1) Restore the equipment configuration and settings to that existing after 2.b(3). The noise for this test will be represented by a single frequency provided by the signal generator.

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(2) Deactivate the Doppler channel for this test by connecting a clip lead from the positive side of C43 to the junction of R52 and R53.
(3) With the attenuators set to 32 ±1db a resistance of 100 ±10 ohms (no alarm state) shall exist across TB1-1 and TB1-2. A resistance of at least 100K ohms (alarm state) shall exist across TB1-1 and TB1-2 when the attenuators are set in the range of 25 ±3db. Allow at least 60 seconds for processor to stabilize when large attenuation steps are made.
(4) Remove clip lead previously installed in 2.c(2).

d. Log Amplifier Gain Criteria.

(1) Configure the equipment as in 2.c(1)
(2) Connect the ac voltmeter to the processor SIG test point and set attenuators to obtain 1.0 ±.1 VRMs at the SIG test point.
(3) Remove 20 db of attenuation; a voltage of 1.8 ±.3 VRMS shall be measured at the SIG test point.

e. Linear amplifier gain and doppler filter criteria.

(1) Configure the equipment as in 2.c(1).
(2) Connect the ac voltmeter to the NOISE test point of the processor and set the attenuators to 20 db. The voltmeter shall indicate 2.8 ±.3 Vrms.
(3) When the attenuation is increased from 20 to 40 db the voltage measured in 2.e (2) shall be reduced by a factor of 10 ±10 percent.
(4) With the attenuators set to 40 db the ac voltage measurement at the NOISE test point shall decrease by at least 3db when the signal generator frequency is set to obtain the following frequency counter measurements at the NOISE test point:

(a) +800 ±20 Hz, Note 1
(b) -800 ±20 Hz, Note 2
(c) +12 ±2 Hz, Note 1
(d) -12 ±2 Hz, Note 2

Notes:
1 - Signal generator frequency set greater than processor frequency measured in paragraph 2.b.(1).
2 - Signal generator frequency set less than processor frequency measured in paragraph 2.b.(1).

f. GATING Waveform Criteria.

(1) Configure the equipment as in 2.c(1)
(2) Connect channel 1 of the oscilloscope to the junction of R51 and R57 and channel 2 to the SIG test point.

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(3) Channel 1 shall indicate a pulse amplitude of +12 ±1 volts. The pulse duration shall be 100 ±35 microseconds in duration.
(4) Switch on Channel 2 of the oscilloscope (dual trace or chopped mode operation) and note that the pulse observed on Channel 1 occurs at the negative peak of the sine wave observed on Channel 2 when the signal generator is set to 400 ±20 Hz greater than that obtained in 2.b (1). Change the signal generator frequency to 400 ± 20 Hz less than that measured in 2.b(1). The pulse observed on Channel 1 shall occur at the positive peak of the sine wave observed on Channel 2.

g. Transmitter failure alarm test.

(1) Configure the equipment as in 2.c(1).
(2) Set the attenuators to 70 db. A resistance of 100 ±10 ohms shall exist between TB1-1 and TB1-2 (no alarm state).
(3) The processor shall generate an alarm condition (at least 100 K ohms between TB1-1 and TB1-2) when the frequency control transducer red lead is disconnected for any duration above 250 milliseconds.
(4) Restoration of the frequency control transducer connection shall result in a processor no alarm condition (100 ohm ±10 ohm between TB1-1 and TB1-2) within 1.5 seconds.

h. Syncronization Output Test.

(1) Configure the equipment as in 2.c(1).
(2) Connect Channel 1 of the oscilloscope to TB2-7.
(3) A symmetrical triangular wave form of 3.0 ±0.3 volts PK to PK shall be observed.

i. Tamper Switch Test.

(1) Depress the tamper switch plunger; a resistance of 1 ohm or less shall occur between TB1-5 and TB1-6.
(2) Pull the tamper switch plunger to the full out position; a resistance of 1 ohm or less shall occur between TB1-5 and TB1-6.
(3) Depress and release the tamper switch plunger; a resistance of at least 100K shall occur between TB1-5 and TB1-6.

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Figure 2, delete and substitute new figure 2.

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After Test Method 8, add Test Methods 9 and 10:

"TEST METHOD 9

TRANSCEIVER, ULTRASONIC MOTION SIGNAL

TA13220E3226

1. TEST APPARATUS:

Same as apparatus of Test Methods 4, 6, and 8.

2. TEST PROCEDURE:

2.1 Transducer tests.

a. Disconnect the leads connecting the transmit and receive transducers to the circuit card assembly.
b. Perform test procedures 2.2 and 2.3 of Test Method 8 on the white dot and blue dot transducers respectively.

2.2 Circuit card tests.

a. With the transducer leads disconnected from the circuit card assembly, perform test procedures 2.1 and 2.2 of Test Method 6.

2.3 Tamper switch tolerance test.

a. Reconnect transducer leads in normal configuration.
b. Perform tamper switch tolerance test in accordance with test procedure 2.2 of Test Method 4."

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APPENDIX

"TEST METHOD 10

PROCESSOR, ULTRASONIC MOTION SIGNAL

(TA 13220E3201)

1. TEST APPARATUS:

Same as apparatus of Test Methods 4 and 7.

2. TEST PROCEDURE:

2.1 Circuit Card Tests.

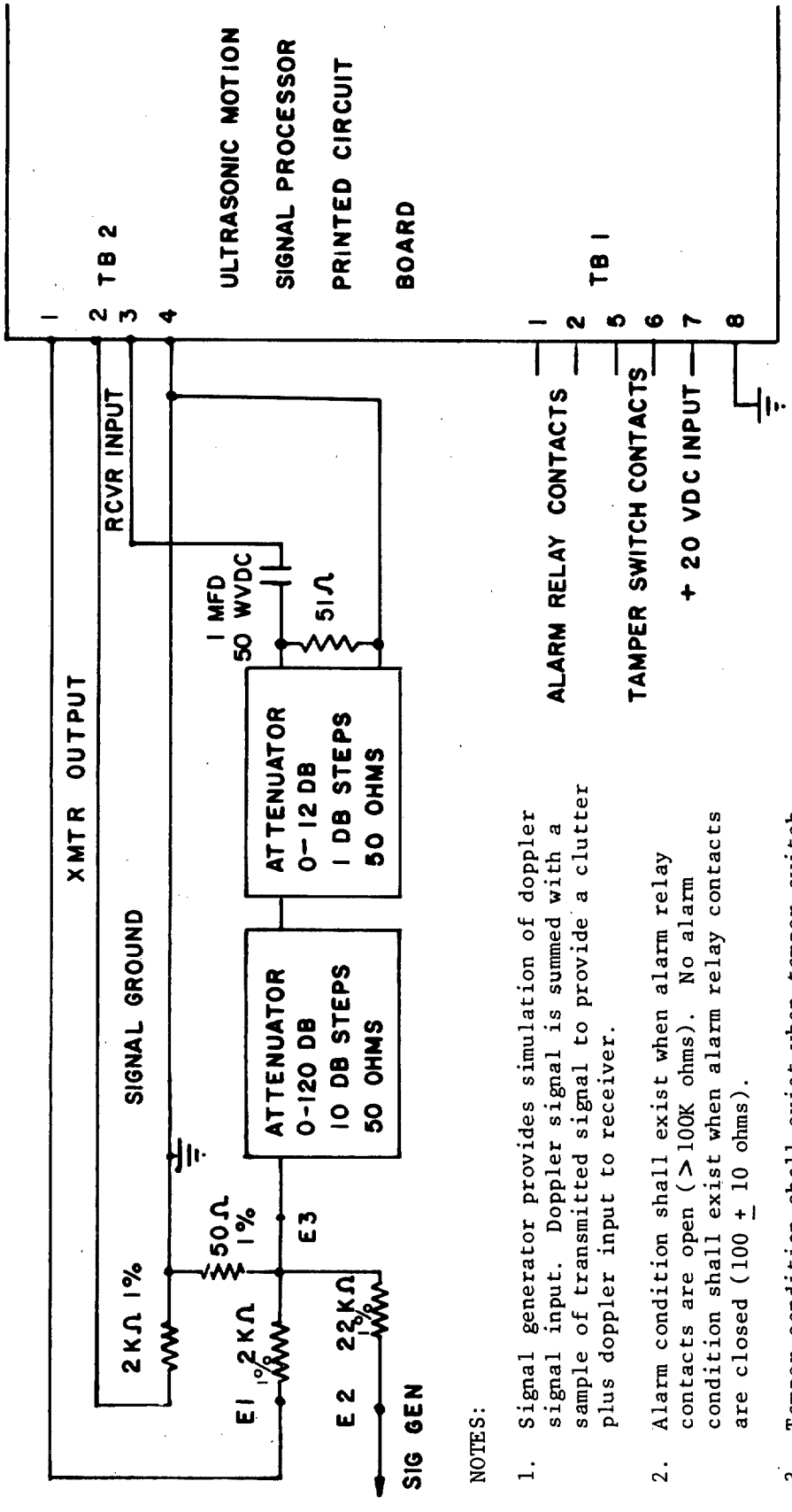
a. Arrange equipment as shown in Test Method 7.
b. Perform test procedures 2a through 2h of Test Method 7.

2.2 Tamper switch tolerance test.

a. Perform test procedure 2.1 of Test Method 4."

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NOTES:

1. Signal generator provides simulation of doppler signal input. Doppler signal is summed with a sample of transmitted signal to provide a clutter plus doppler input to receiver.
2. Alarm condition shall exist when alarm relay contacts are open (>100K ohms). No alarm condition shall exist when alarm relay contacts are closed (100 ± 10 ohms).
3. Tamper condition shall exist when tamper switch contacts are open (>100K ohms). Non-tamper condition shall exist when tamper switch contacts are closed (<1 ohm).

FIGURE 2. Test fixture for ultrasonic motion signal processor printed circuit board.

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